

Compost: The Garden Stream to the Beds

Carpathian OER Commons — Land, Forest, Food Module

Michel Garand | 2026-09-28 | Version v0.1 | CC BY-SA 4.0

Maturity: L0, a marked draft — sized from the market garden's need and from what the site can supply, with the bought-in compost priced from a dated listing, and nothing yet made. Transport, the base under the bays, and the covers are not priced; no pile has been built and no distance has been measured on the ground. Read every figure below at this level and no higher. The module moves to L1 when every line in Section 6 carries a dated cost and its basis.

Governed by Section 2 of the compendium. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only.

Two streams, kept apart. The toilet stream — buckets, biochar, sealed lactic fermentation, worms, a year's maturation — goes to trees only and belongs to the terra preta module. This module is the other one: the garden's own stream, from kitchen, garden, and manure to the beds. Decided 2026-09-28.

1. Purpose

Compost for the market garden's beds — about 62 m³ to build them and about 22 m³ a year after — from the garden's own stream, and bought in for the share the site cannot make. It says how much the site can make, how much must be bought, for how long, and where the bays may stand.

Five rules. Two streams kept apart: nothing from the toilets, and no cooked food, plate waste, or grease-trap solids, reaches the garden pile — those go to the toilet stream's sealed vessels and to trees. Charged char only: char enters the pile as it is built, 10 to 30 percent by volume, and never goes raw onto a bed. Hot is logged, not assumed: a pile counts as hot only when its thermometer record meets the sanitising steps in Section 7; every other pile is cold compost, for edges, paths, and perennials. Distance before convenience: bays, piles, and the manure heap keep the distances in Section 4, measured on the ground, and the fallback is a longer carry, never a shorter distance. Compost never leaves the hamlet: it is made and used here, and neither sold nor given away. The work is the grower's, paid; toloka is not the name for work the compost depends on.

Connections. Depends on: the market garden module for the need — its bed area, its compost depths, and its rotation; the Phase 1 walk, with the spring, the stream, the garden's pond, and every dwelling and place where children are marked, and the distances in Section 4 measured with a tape; the biochar module for charged char, a second exit from the same kiln; the kitchen and dining module for raw scraps; a dated local quotation for rotted manure or compost; the household's decision on animals. Feeds: the market garden module, with finished compost to the beds and a price for the compost it buys. Hands to the terra preta module: cooked food, plate waste, and grease-trap solids, which that module sizes into its vessels.

Patterns. Grows from: not yet named. How the valley bedded its animals and fed its gardens, where the manure heap stood relative to the house, the spring, and the garden, and whether anything like a compost heap was kept apart from it, are questions for the elders before the first bay is built. Feeds: not yet named. Resonates with: 177 Vegetable Garden — the bays beside the plots they feed; 104 Site Repair — bays, heap, and char store on the poorer ground; 25 Access to Water — the protective strip keeps every pile out of the belt along the stream and the pond; 64 Pools and Streams — covered bays on a firm base keep leachate from running water; 169 Terraced Slope — a bay built level on a terrace; 120 Paths and Goals — the bays on the path the kitchen already walks. In tension with: 178 Compost — Alexander joins toilets and kitchen waste into one heap; here the toilet stream goes to trees and only the garden stream reaches the beds, and cooked food goes with the toilet stream. 74 Animals — animals among people, where the law keeps a manure heap 20 m from any dwelling and from places where children are. 172 Garden Growing Wild — cold compost feeds the edges and is kept off the seed beds.

2. Context of validity

Condition	Range this design holds for
Climate	[set] — mountain; piles freeze in winter
Warm season	frost_free_days, [set]; hot piles built in it
Altitude	[set]
Need	the market garden's 1,440 m ² of bed
Group size	capacity: 34 people, the example below
Days of demand	use_days_per_year: 120, the example below
Animals	none kept, the example; Section 3 adds them
Ground	a level, firm place outside every distance

The need is the market garden's, at its own compost depths, taken from Fortier (The Market Gardener, 2014, pp. 45, 56). A smaller garden, or the same garden built in stages, needs less; Section 3 shows both.

3. Sizing method

The need is fixed by the beds. The supply is what the site's own stream can make. The difference is bought. All three are read in cubic metres of finished compost.

What you must measure yourself before using this module:

- **Your need** — the bed area and the compost depths you mean to use, from your own garden plan.
- **Your kitchen's raw scraps** — weighed for a full cooking week, raw plant material only, kept apart from cooked food and plate waste.
- **Your garden's residues** — what leaves the beds for the pile, not what stays under the tarp, weighed at each clearing, from the bed record.
- **Your manure** — the animals you keep or can take from, their stall days, and the bedding used.
- **Your distances** — spring, stream, pond, dwellings, kitchen, and places where children are, measured with a tape from the proposed bay, on the walk.
- **Your price** — a dated quotation for rotted manure or compost, delivered, from the nearest supplier.

Our example: capacity, 34 people; use_days_per_year, 120; vegetables_per_person, 0.44 kg; garden_share, 1.0; prep_refuse_share, 7 to 31 percent; residue_to_pile, 0.2, [assumed]; compost_volume_ratio, 0.33 to 0.75; rotted_manure_density, 900 kg per m³. Values from the site parameters file, held in the workshop. The bed area, 1,440 m², and the compost depths are the market garden module's.

3.1 The need

Quantity	Calculation	Our example
To build the beds	1,440 m ² × 4.3 cm	about 62 m ³
Each year after	720 m ² × 3.0 cm	about 22 m ³
Built in stages, 60 beds first	900 m ² × 4.3 cm	about 39 m ³
The other 36 beds, a year later	540 m ² × 4.3 cm	about 23 m ³

Sixty beds are what the kitchen's demand needs in the market garden's own arithmetic; the other thirty-six carry the margin. Building them first spreads the purchase over two years and changes nothing else.

3.2 What the site can make

Stream	Calculation	Raw, a year
Kitchen, raw scraps	1,795 kg served × refuse	135–806 kg
Same, by volume	÷ 275 kg per m ³	0.5–2.9 m ³
Garden residues, all	1,440 m ² × 1.0–5.5 kg	1,440–7,920 kg

Stream	Calculation	Raw, a year
Carried to the pile	$\times \text{residue_to_pile}, 0.2$	288–1,584 kg
Same, by volume	$\div 138\text{--}148 \text{ kg per m}^3$	1.9–11.5 m ³
Leaves, yard and trees	counted on the walk	[set]
Manure	no animals, the example	none

Finished compost	Calculation	Our example
From the kitchen	$0.5\text{--}2.9 \text{ m}^3 \times 0.33\text{--}0.75$	0.2–2.2 m ³
From the garden	$1.9\text{--}11.5 \text{ m}^3 \times 0.33\text{--}0.75$	0.6–8.6 m ³
Site-made, a year	sum	0.8–10.8 m³

The kitchen. The 1,795 kg is the vegetables the kitchen serves in a year at capacity, net, from the market garden's demand line. What is peeled and trimmed away before serving is added to it: prep_refuse_share runs from 7 percent of the purchased weight for onion to 31 percent for headed cabbage in FAO's tables (Chatfield, Food composition tables for international use, 1953, Table 2), and from 12 to 30 percent in the USDA's yields for school kitchens (Food Buying Guide, Section 2). Net to gross: refuse = net $\times$ share $\div$ (1 – share). Food waste is taken at 275 kg per m³ (US EPA, Volume-to-Weight Conversion Factors, 2016). The resident households' scraps are not yet counted, [set].

The garden. The residue per m² runs from 1.0 kg for head lettuce to 5.5 kg for zucchini, fresh, in Lower Saxony's advisory values (Landwirtschaftskammer Niedersachsen, 2007, Table 2). Most of it never reaches the pile: the market garden cuts residues and covers them with a black tarp on the bed, as Fortier does (Section 7 of that module), and they feed the bed where they lie. residue_to_pile, 0.2, is the share that leaves the bed — the bulky, the diseased, the tunnel's haulm — and is [assumed] until the bed record gives it. Green garden waste is taken at 138 to 148 kg per m³ loose (Minnesota PCA, 2021; US EPA, 2016).

The shrink. compost_volume_ratio is the finished volume per raw volume. The sources disagree: NC State gives a fall of 25 to 50 percent (Composting on Organic Farms, 2006); the US EPA a pile that shrinks to about one-third (Composting at Home); NDSU a fall of half to two-thirds for manure (2011); Rynk a fall of one-quarter to more than one-half (On-Farm Composting Handbook, 1992, p. 12). The range, 0.33 to 0.75, is carried through; the site's own piles replace it.

Char is not counted. Charged char, 10 to 30 percent of the pile by volume (Schmidt, Ithaka Institute), adds structure and holds nutrients, and the finished pile is larger by that much. It is left out of the supply above, because it feeds nothing on its own, and because how much the kiln can give beyond the toilet stream's 1.5 m³ a year is the biochar module's to say, [set].

3.3 What must be bought

Year	Need	Site-made	Bought
Building the beds	62 m ³	none finished yet	about 62 m ³
Each year after	22 m ³	0.8–10.8 m ³	about 11–21 m ³

In plain words. The beds are built from bought compost or rotted manure, all of it: nothing the site makes is finished before the first beds are dug. After that, the site makes between a twenty-fifth and a half of what the beds take, and the rest is bought — every year, for as long as the garden runs at this size and the household keeps no animals. It is a running cost, not a start-up cost. The gap closes only by keeping animals, by taking a neighbour's manure, or by fewer beds. Fortier advises beginning growers to buy their compost rather than make it (p. 62), and buys his own from specialists (Kaminski, review in Resilience, 2014).

3.4 If the household keeps animals

Animal	Bedded manure, a year	Rotted, as m ³
One cow, 200 stall days	about 7 t	about 1.6–4.7 m ³
One sheep	about 0.5 t	about 0.1–0.3 m ³

Bedded manure per head is from a Ukrainian agrochemistry textbook: about 7 t for a cow with 2 kg of straw a day over 200 stall days, about 0.5 t for a sheep (Арпохія, section on manure); a Vinnytsia agrarian university guide gives cattle 4 to 13 t and sheep 0.4 to 1.2 t by the length of the stall period. Rotting leaves 20 to 60 percent of the mass (Rynk, 1992, p. 12); перерний weighs about 900 kg per m³ (Бондарчук, Lutsk NTU, practical class 2). A cow and a few sheep would close a tenth to a half of the yearly gap; they would not close it. The stall days in this valley are [set], and are for the household and the elders [for the elders].

4. Bill of materials

Quantities for the example. The three bays are the market garden's, $3 \times 3 \times 1.2$ m, costed there; they hold about 32 m³, enough for a year's bought compost stacked to mature beside the site's own piles.

Item	Quantity	Alternatives in the design
Rotted manure, to build the beds	about 62 m ³	compost; two stages
Rotted manure, each year after	about 11–21 m ³	compost
Compost thermometer, 41 cm probe	1	none named
Rodent mesh under the bays	30 m ²	none named
Compost fork	1	none named
Covers for the bays	3, [set]	boards; old tunnel film
Firm base under the bays	27 m ² , [set]	clay, rammed
Kitchen caddies, lidded	[set]	none named

The siting, as designed. Every distance below is the larger of what the instruments in Section 10 ask, and every one is measured on the ground before a bay is built; none has been measured yet.

From the bays, heap, and piles to	At least
The spring, and never upslope of it	50 m
The stream, on a slope over 3°	50 m
The garden's pond, on a slope over 3°	50 m
Any dwelling or summer kitchen	20 m
Any place where children are	20 m
A private well, if there is one	20 m

The Water Code keeps vegetable beds out of the same strip along the stream and the pond, so the bays can stand beside the plots only where the plots themselves stand clear of it. If the ground left between these distances is too narrow for bays beside the plots, the bays stay with the plots and the kitchen's scraps are carried farther.

The layout, as designed. The bays beside the plots, on the poorer ground, level, on a firm base, covered, on the path from the kitchen; the manure heap beside them, never nearer the water; the char store with the kiln, as the biochar module places it. Drawn as relationships only; the walk sets the rest.

5. Build

Step	Trade	Time	If done out of order
Walk; distances measured	grower	[from L2]	bays in the strip
Quotation for manure, dated	grower	[from L2]	cost unknown
Base levelled and firmed	grower	[from L2]	leachate runs off
Rodent mesh laid	grower	[from L2]	[from L2]
Bays built, market garden	carpenter	[from L2]	[from L2]
Manure delivered, checked	supplier	[from L2]	raw manure on beds
First beds built	grower	[from L2]	[from L2]

Bought manure is checked on delivery: well rotted, crumbly, without smell of ammonia. Manure that is not goes into a bay for six months or more before it reaches a bed, as Dowding advises; bought compost is stacked up to three months to mature (Dowding, No Dig FAQs). A bean test before spreading shows herbicide residue (Section 8).

6. Cost

Estimates, not quotations: each line is a price listed by a Ukrainian retailer or service, seen on 2026-09-28. Place: not a quotation; the manure listing is from the Kyiv region, since no bulk listing was found in western Ukraine. Exchange rate: exchange_rate, 52 UAH to the euro, the National Bank's mid-September 2026 rate, rounded.

6.1 Building the beds

Item	UAH	EUR	Basis
Rotted manure, 62 m ³ , about 55.8 t	39,060	751	estimated, listing
Transport from the supplier	[set]	[set]	not priced

Item	UAH	EUR	Basis
Subtotal, priced lines	39,060	751	

The manure line is 700 UAH a tonne for cattle manure, minimum 5 t, transport extra, Kyiv region, seen 2026-09-28; a second listing there gives 600 UAH a tonne. Tonnes from m³ at rotted_manure_density. Built in two stages, the first 39 m³ is about 35.1 t, 24,570 UAH.

6.2 Tools and bays

Item	UAH	EUR	Basis
Compost thermometer, 41 cm probe	835	16	estimated, listing
Rodent mesh, 30 m ² at 117.30 UAH	3,519	68	estimated, listing
Compost fork	599	12	estimated, listing
Covers for the bays	[set]	[set]	not priced
Firm base under the bays	[set]	[set]	not priced
Kitchen caddies	[set]	[set]	not priced
Subtotal, priced lines	4,953	95	

6.3 Summary

Section	UAH	EUR
Building the beds, priced lines	39,060	751
Tools and bays, priced lines	4,953	95
Construction subtotal	44,013	846
Design check, 8 percent	3,500	67
Contingency, 20 percent	8,800	169
Total external cost, priced lines	56,313	1,083

The total leaves out transport, the covers, the base, and the caddies. It is not the module's cost until they are priced. The percentages are the working papers' own. The bays themselves are in the market garden's total, not here.

Contribution, not in the total. None counted. The grower's work is a cost, not a contribution.

Running cost. The bought compost, 11 to 21 m³ a year, about 10.1 to 19.1 t at the same listing: 7,056 to 13,356 UAH a year, transport not included — the largest recurring line the garden's ground has, and the line the market garden has been waiting for. The grower's time at the piles is inside the grower's pay, set in the market garden module.

7. Operation and maintenance

Responsible: the grower, with a deputy — [from L2]. One named role, paid; not a rota of residents.

- **Daily, in season:** raw kitchen scraps carried from the kitchen in lidded caddies and covered in the bay with dry material; cooked food, plate waste, and grease-trap solids never go to the bay — they go to the toilet stream's vessels.
- **As each pile is built:** greens and browns layered, about 30 parts carbon to 1 of nitrogen and about half water by weight, the ranges the University of California gives (Compost in a Hurry, 2001); charged char added through the pile; at least 0.9 m in each direction, the University of California's 36 inches, so that it can heat.
- **While a pile is hot:** the temperature read and written down each day at its centre. A pile counts as hot only when the record shows at least 3 days at 70 °C, 5 at 65 °C, 7 at 60 °C, or 14 at 55 °C — the sanitising steps of Regulation (EU) 2019/1009 — with turning to bring the outside in. Otherwise it is cold compost.
- **Seasonal:** hot piles built in the warm months; piles covered against rain and snow; the manure heap covered; bought manure stacked on arrival.
- **Annual:** the compost record — what went in, when, how hot, what came out — carried into the bed record and into prep_refuse_share, residue_to_pile, and compost_volume_ratio; the distances walked again.

Handling is by the grower and paid help. The toilet stream and its open vessels belong to the terra preta module, whose rules hold for them. If the programme works in the garden, that is described in the programme modules, in standing format.

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Pile never heats	record under 55 °C	cold compost, edges only

Failure	Early sign	Repair and cost
Leachate toward water	[from L2]	cover, base, distance
Rats in the bays	[from L2]	mesh; no cooked food
Weed seed in bought manure	[from L2]	stack; a hot pile
Herbicide in bought manure	bean test fails	refuse the load
Raw manure on the beds	[from L2]	120 days before harvest
Compost short of the beds	[from L2]	buy more, or fewer beds
Supplier too far to pay	[from L2]	a nearer source

Herbicide in bought manure. Some grassland herbicides, aminopyralid among them, pass through the animal and the heap and damage beans, peas, potatoes, and tomatoes grown in the manure; the RHS advises a simple test — beans sown in pots of the manure beside pots without — before using manure of unknown origin (RHS, Weedkiller in manure). The supplier is asked what the animals were fed.

Raw manure. No Ukrainian interval between raw manure and harvest was found. The design takes 120 days before harvesting a crop that touches the soil, and 90 for one that does not, from the US federal rule on manure in crop production (7 CFR 205.203), [set], and keeps raw manure off the beds altogether.

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	found 2026-09-28, not checked — see below	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—

Country	Instrument, number, date	Checked by, date
Serbia	not checked	—
Czechia	not checked	—

Ukraine, found and not checked. Instruments found on 2026-09-28 and read on secondary copies and official summaries, because the statute site could not be reached; each is read on zakon.rada.gov.ua and signed before this row counts as checked.

- **Water Code, No. 213/95-BP of 06.06.1995.** Art. 88: a coastal protective strip of 25 m along small streams and ponds under 3 ha, doubled where the slope exceeds 3°. Art. 89: inside it, no vegetable growing, no storing or applying fertiliser, no manure store.
- **ДСанПіІН 2.2.4-171-10, МОН Order No. 400 of 12.05.2010.** Point 3.32: a spring capture at least 50 m from fertiliser stores, places where livestock are kept, and latrines; 20 m for a private well.
- **Cabinet Resolution No. 2024 of 18.12.1998,** on sanitary protection zones: a first belt of 30 or 50 m, formally for centralised supply; whether it is asked of a private spring serving a children's establishment is for the enquiry.
- **ДБН Б.2.2-12:2019.** Compost sites and manure stores at least 20 m from dwellings, summer kitchens, and drinking wells; the clause number is not verified.
- **МОН Order No. 377 of 20.03.2026,** the sanitary regulation for children's health and recreation establishments: waste containers at least 20 m from places where children are; nothing on compost or manure was found in its summaries, and its full text is to be read.
- **Order No. 489 of 13.06.2023, Minjust No. 1271/40327,** the rules for composting biowaste on household and garden plots: no permit; at least 5 m from a private well and 50 m from a public water source; no meat, fish, bones, dairy, fats, or animal faeces; compost for the generator's own needs. Whether it covers a garden that is a registered capacity of an operator is uncertain.
- **Law No. 287-VIII of 07.04.2015,** on animal by-products: manure is Category II (Art. 12) and may be composted or applied to soil (Art. 15, as revised by Law No. 1206-IX of 2021).
- **Law No. 2320-IX of 20.06.2022,** on waste management: by-products that are composted fall inside it (Art. 2); no end-of-waste criteria for compost have been adopted (Art. 8).
- **Law No. 86/95-BP of 02.03.1995,** on pesticides and agrochemicals: organic fertilisers are agrochemicals (Art. 1), and production, trade, and application need state registration (Art. 4); no exemption for compost made on the farm was found. The module sells and gives away no compost.

The written enquiry to the district food-safety service, with the design drawn, asks four things: the distances that bind this site; whether a first belt is set around the spring; whether Order No. 489 covers the garden; and whether Art. 4 of Law No. 86/95-BP asks anything of compost made and used on the same ground.

11. Adaptation notes

The design is a seed, not a copy: a different place grows a different form from it.

- **A drier valley:** a pile dries before it heats; the covers matter more, and the pile is watered from the pond, never the spring.
 - **A colder valley:** a shorter warm season means fewer hot piles and more cold compost; bought compost carries more of the beds.
 - **A smaller group or less ground:** fewer beds, less need; the kitchen's share of the supply rises, and the bought share falls.
 - **A larger group or more ground:** more beds, more need; the gap grows faster than the kitchen's scraps.
 - **Animals kept:** manure becomes the largest stream; bedding and stall days decide it, and the heap keeps its distances.
 - **Level ground far from water:** the siting table relaxes nothing, but is easy to meet; the bays stand where the path already goes.
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12. Provenance and licence

- **Written by:** Michel Garand. [set — others as they choose to be named]
- **From whose knowledge:** Jean-Martin Fortier, *The Market Gardener*, 2014, for the need, by page; Albert Howard, *The Waste Products of Agriculture*, 1931, with Y. D. Wad, and *An Agricultural Testament*, 1940, for the heap; the University of California's *Compost in a Hurry*, 2001, for the hot pile; Regulation (EU) 2019/1009 for the sanitising steps; Charles Dowding for maturing bought manure and compost; Bruno Glaser and colleagues, 2001, and Claudia Kammann and colleagues, 2015, and Hans-Peter Schmidt of the Ithaka Institute, for charged char; FAO, the USDA, the US EPA, the Minnesota PCA, NC State, NDSU, Robert Rynk, and Landwirtschaftskammer Niedersachsen for the supply figures; a Ukrainian agrochemistry textbook, a Vinnytsia agrarian university guide, and Lutsk NTU for manure; the RHS for the bean test; Ukrainian retail listings of 2026-09-28 for prices.
- **Traditional knowledge:** none used. The literature records the moving of the sheepfold across the polonyna to manure it, the wintering of cattle at the distant hayfield, and fallen leaves and needles as bedding [for the elders]; none of it enters the design until the elders have confirmed, corrected, or withheld it.
- **Text:** CC BY-SA 4.0
- **Drawings and designs:** [hardware licence — open decision]
- **Measurement data:** none yet

- **Gate 1:** household decision, 2026-09-29, recorded by Michel Garand — v0.1 may be published as it stands, as a marked draft, in English, Ukrainian, and German
 - **Wartime review:** 2026-09-29, Michel Garand — v0.1 read against compendium Section 2; nothing places the site; no supplier, market, or town near the site is named; the distances in Section 4 are the instruments' minimums, not measurements of the site; accepted
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13. What to check

- **The distances** — every row of the siting table, measured with a tape on the walk from the proposed bays: spring, stream, pond, dwellings, summer kitchen, places where children are, any well; the slope beside the stream and the pond, to settle 25 or 50 m.
 - **A local price** — a dated quotation for rotted manure or compost, delivered, from the nearest supplier in western Ukraine, replacing the Kyiv-region listing; and the transport.
 - **prep_refuse_share** — the kitchen's raw scraps weighed for a cooking week.
 - **residue_to_pile** — the bed record: what leaves the beds for the pile.
 - **compost_volume_ratio** — the site's own piles, measured in and out.
 - **Leaves** — the yard's trees and fruit trees counted on the walk.
 - **Char** — how much the kiln can give the garden beyond the toilet stream, from the biochar module.
 - **Animals** — the household's decision; stall days in this valley.
 - **The raw-manure interval** — a Ukrainian rule, if one exists.
 - **The Ukrainian row** — every instrument read and signed; the written enquiry.
 - **Covers, base, caddies** — priced.
 - **For the terra preta module** — cooked food, plate waste, and grease-trap solids added to its vessels, sized there.
 - **For the market garden module** — the compost price in its Section 6.1, and its beds, bays, and pond kept out of the Water Code strip, in its next version.
 - **From L2, once built:** build times and out-of-order consequences in Section 5; the grower's daily tasks in Section 7; early signs and repair costs in Section 8; Section 9.
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